INRC-II × Contextual Bandits: Project Proposal

1. Problem Overview

INRC-II (Second International Nurse Rostering Competition) is a benchmark optimization challenge in healthcare scheduling. The core task is to assign nurses to shifts (Early, Late, Night) or days off across every day of a week — and to repeat this decision-making process across a planning horizon of **4 or 8 consecutive weeks**.

Unlike its predecessor (INRC-I), INRC-II uses a **multi-stage formulation**: decisions made in one week carry consequences into future weeks via a shared **history** state. The objective is to minimize the total soft constraint penalty across the entire planning horizon while never violating hard constraints.

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2. Data Sources

Each week, the solver receives input from three sources:

2.1 Scenario (global, fixed across all weeks)

- Planning horizon length (4 or 8 weeks)
- Nurse roster: names, contracts, skills
- Contract rules: min/max total assignments, min/max consecutive working days, min/max consecutive days off, max working weekends, complete weekend constraint
- Shift types and forbidden shift successions (e.g., Night → Early is illegal)

2.2 Week Data (changes each week)

- Coverage requirements: optimal and minimum nurses needed per shift, per skill, per day
- Nurse day-off requests for the week

2.3 History (passed between weeks — the rolling state)

The history file encodes each nurse's current cumulative state. Example:

```
json
```

```
{
  "week": 0,
  "scenario": "n030w4",
  "nurseHistory": [
    {
      "nurse": "HN_0",
      "numberOfAssignments": 0,
      "numberOfWorkingWeekends": 0,
      "lastAssignedShiftType": "Night",
      "numberOfConsecutiveAssignments": 4,
      "numberOfConsecutiveWorkingDays": 4,
      "numberOfConsecutiveDaysOff": 0
    },
    {
      "nurse": "HN_1",
      "numberOfAssignments": 0,
      "numberOfWorkingWeekends": 0,
      "lastAssignedShiftType": "None",
      "numberOfConsecutiveAssignments": 0,
      "numberOfConsecutiveWorkingDays": 0,
      "numberOfConsecutiveDaysOff": 5
    }
  ]
}
```


Key fields and what they mean:

Field	Significance
`lastAssignedShiftType`	Determines legal shift successions for Monday
`numberOfConsecutiveAssignments`	Risk of violating max consecutive days rule
`numberOfConsecutiveDaysOff`	Risk of violating min consecutive days off rule
`numberOfWorkingWeekends`	Proximity to contract weekend cap
`numberOfAssignments`	Proximity to contract total assignment limits

> Example: HN_0 just worked 4 consecutive nights and has 0 days off — scheduling another night shift immediately may be a hard constraint violation. HN_1 has been off 5 days and is fresh.

3. Constraint Structure

Hard Constraints (never violate)

- Minimum staffing coverage per shift/skill/day
- Forbidden shift-to-shift successions
- Contract limits on consecutive working days and days off

Soft Constraints (minimize total penalty)

- Optimal (above-minimum) staffing levels
- Maximum consecutive working days / days off
- Maximum working weekends over the planning horizon
- Complete weekend constraint (work both Saturday and Sunday, or neither)
- Nurse day-off preferences

The **final penalty score** is computed by the INRC-II validator after all weeks are concatenated. Many soft constraints (e.g., total assignments, total working weekends) only fully crystallize at the **end** of the horizon.

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4. Why This Problem Is Hard

1. **Sequential coupling**: A locally good schedule in week 1 can make week 3 infeasible or very costly due to accumulated history state.
2. **Delayed rewards**: Many soft constraints are only fully evaluated at the end of the planning horizon.
3. **Heterogeneous agents**: Nurses have different contracts, skills, and preferences — they are not interchangeable.
4. **Combinatorial scale**: With 30 nurses × 7 days × 4 shift options, the roster space is astronomically large.
5. **Non-stationary context**: Each week presents different demand patterns and nurse states.

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5. Applying Contextual Bandits

5.1 Framework Mapping

Bandit Component	INRC-II Mapping
Context	History file + week data (nurse states, coverage needs, week number)
Action (arm)	A roster-generating heuristic policy
Reward	Negative soft constraint penalty for the week (proxy) or end-of-horizon score
Episode	One planning horizon (4 or 8 weeks)

5.2 Context Vector

For each nurse, extract features from the history file:

- `lastAssignedShiftType` (one-hot encoded)
- `numberOfConsecutiveAssignments` (normalized by contract max)
- `numberOfConsecutiveDaysOff` (normalized by contract min)
- `numberOfWorkingWeekends / contractMaxWeekends`
- `numberOfAssignments / contractMaxAssignments`

From the week data, add aggregate features:

- Total coverage demand per shift type across the week
- Number of nurse day-off requests
- Week number / total weeks (position in horizon)
- Aggregate nurse "fatigue" signal (e.g., mean consecutive working days across nurses)

5.3 Action Space: Policy Selection

Rather than treating each possible roster as an arm (intractable), define a small set of **deterministic heuristic policies** as arms. The bandit learns *which policy to deploy* given the week's context.

Example candidate policies (arms):

1. **Coverage-first greedy**: Prioritize filling minimum staffing requirements before any other objective.

2. **Fatigue-aware**: Prioritize assigning rest to nurses with high consecutive working day counts.
3. **Weekend-balance**: Minimize the variance of working weekends across nurses.
4. **Preference-respect**: Maximize honoring nurse day-off requests when coverage allows.
5. **Lookahead-conservative**: Assign schedules that preserve flexibility for future weeks (e.g., avoid using up weekend budget early).

The bandit selects one arm per week, observes the resulting penalty, and updates its policy.

5.4 Reward Signal

The natural reward is the **negative soft constraint penalty** for the produced roster. However, this is complicated by delayed feedback:

- **Per-week proxy reward**: Penalize only the constraint violations that can be evaluated within the current week (e.g., shift successions, coverage shortfalls, preference violations). Use this as the immediate signal.
- **End-of-horizon adjustment**: At the end of the planning horizon, compute the full penalty and use the difference as a corrective signal for the final week's arm choice.
- **Shaped reward**: Linearly interpolate between the per-week proxy and an estimated future cost based on current nurse state margins.

5.5 Bandit Algorithm Options

Algorithm	Suitability
LinUCB	Strong baseline; assumes linear reward in context; fast and interpretable
Neural / Deep Contextual Bandit	Captures non-linear context relationships; requires more data
Thompson Sampling (contextual)	Good exploration-exploitation balance; Bayesian posterior over reward

LinUCB is recommended as the starting point given the relatively small number of arms and the need for interpretability.

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6. Project Objectives

1. **Formalize the contextual bandit framing** of INRC-II and justify it against alternative approaches (full RL, static optimization).
2. **Engineer a context vector** from history JSON and week data files across INRC-II benchmark instances.
3. **Define and implement arm heuristics** (at minimum 4–5 policies) that produce feasible weekly rosters.
4. **Design a shaped per-week reward** that correlates with final horizon penalty and handles delayed feedback.
5. **Train a contextual bandit** (LinUCB as baseline) across multiple INRC-II scenario instances and evaluate generalization.
6. **Benchmark** bandit-selected heuristic sequences against: (a) always using the single best heuristic, (b) random arm selection, and (c) published INRC-II benchmark scores where comparable.

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7. Key Challenges and Mitigations

Challenge	Mitigation
Delayed/sparse rewards	Proxy per-week reward + end-of-horizon correction
Hard constraint feasibility	Feasibility filter or repair operator applied before committing any roster
Small interaction budget (4–8 weeks per episode)	Train across many INRC-II scenario instances; treat each as a separate bandit episode
Action space tractability	Policy selection framing (fixed set of heuristic arms) rather than full roster enumeration
Non-stationarity across weeks	Contextual bandit naturally handles this; context encodes the week-specific state

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8. What Success Looks Like

- The bandit learns that **coverage-first** is best in early weeks with high demand, but **fatigue-aware** is better in later weeks when consecutive assignment counts are elevated.
- The bandit outperforms any single fixed heuristic applied uniformly across all weeks.
- The bandit generalizes across different INRC-II scenario instances (different nurse counts, different horizon lengths).

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9. References

- Ceschia, S., Dang, N. T. T., De Causmaecker, P., Haspeslagh, S., & Schaerf, A. (2015). *Second International Nurse Rostering Competition (INRC-II) — Problem Description and Rules.* arXiv:1501.04177.
- Ceschia, S., & Schaerf, A. (2020). *Solving the static INRC-II nurse rostering problem by simulated annealing based on large neighborhoods.* Annals of Operations Research.
- INRC-II Competition Website: http://mobiz.vives.be/inrc2/